

DHRITIKAMAL DAS

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PROFILE

Machine Learning Engineer specializing in Generative AI, RAG architectures, and computer vision pipelines. Proven track record of developing production-ready AI systems with measurable retrieval and classification accuracy.

CORE SKILLS

- **Generative AI** : LLMs, Generative AI, RAG, Prompt Engineering, OpenAI API, Embeddings, Vector Search, Hybrid Search, Reranking
- **AI/ML** : PyTorch, Scikit-learn, XGBoost, Machine Learning, Computer Vision, Model Evaluation, Feature Engineering
- **LLM Frameworks** : LangChain, LlamaIndex
- **Programming & APIs** : Python, FastAPI, REST APIs, JSON
- **Databases**: PostgreSQL (Supabase), MySQL, FAISS
- **MLOps & Infrastructure**: Docker, CI/CD, Git, GitHub, GitHub Actions
- **Data**: Pandas, NumPy, SQL

PROFESSIONAL EXPERIENCE

JYESTA Corporate Entity | Machine Learning Intern | 10-May-2026 - 10-Jul-2026

- Developed predictive machine learning models using Python.
- Cleaned complex real-world datasets for training.
- Conducted statistical data preprocessing and feature engineering.
- Evaluated model performance metrics against business requirements.

KEY PROJECTS

AI Research Assistant — RAG & LLM Application

Live Demo: <https://ai-research-assistant-cjckybbelkgw89ghmdy53.streamlit.app/>

GitHub Code: <https://github.com/Dhritikamal191/AI-Research-Assistant>

- Engineered an enterprise-grade document analysis solution using Llama 3.
- Achieved 80.13% faithfulness across validation test scenarios.
- Secured 79.12% answer relevancy using automated benchmarks.
- Combined FAISS and BM25 for hybrid retrieval pipelines.
- Implemented cross-encoder reranking to optimize context retrieval.
- Built conversational memory and source citation into production.
- Integrated response caching and user feedback logging systems.

Medical Image Classification

Live Demo: <https://ai-knee-detection-project-tux8wz65q4hr2nz47jdzew.streamlit.app/>

GitHub Code: <https://github.com/Dhritikamal191/AI-Knee-Detection-Project>

- Built an Ordinal ResNet50 computer vision pipeline.
- Classified 5-level knee osteoarthritis with 60.93% accuracy.
- Maintained a 59.89% Macro F1-score during training.
- Integrated Grad-CAM explanations for visual model transparency.
- Wrote 11 automated pytest test scripts for validation.

EDUCATION

Indira Gandhi National Open University (IGNOU)

M.Sc(Mathematics with Application in Computer Science)(MSCMACS)

Jan-2024 - Jun-2026

B.H College (under Gauhati University)

B.Sc(Mathematics Hons.) : 7.23 CGPA

Oct-2020 - Jun-2023